

The FIC Paradigm of Digital Optical All-Optical Computing

(The FIC Paradigm: Axioms, Periodic Table, and Physical Anchors)

Manifesto: The Dawn of Digital Optics

For the past three decades, global photonics research has attempted to clumsily simulate "electronics (current/voltage)" using "wave optics (interference/diffraction)", ultimately falling into an inescapable abyss of cascading collapse and noise disasters. Traditional all-optical computing treats light as an "analog wave", making it impossible to bridge the chasm from "functional devices" to "general-purpose computers".

The **FIC (Photonic State Constructor)** architecture declares the end of this obsolete paradigm. FIC refuses to clumsily simulate electronics with optics; instead, it returns to the fundamental physical nature of photons: **elevating the underlying logic of optical computing from "wave interference" to "constraint and projection in state space"**.

This framework introduces the physical paradigm of "**Digital Optics**" for the first time: abandoning fragile dark states and analog interference, and utilizing exquisitely designed **Physical Mapping Configurations** (M_C) to construct **Photonic State Constructor operators** (PSC). By allowing light to naturally collapse into energy-rich **Orthogonal Bright States** within "attractor potential wells", FIC achieves all-optical digital computing with inherent self-healing and infinite cascading capabilities.

This is not merely a device innovation; it is the **physical constitution** for the next generation of all-optical computing.

Part I: The Axioms of Physical Digital States

Before assembling any logic, the FIC framework must first establish three unshakeable physical axioms to redefine the "digital states of light".

Axiom I: The Axiom of Orthogonal Bright States

- **The Traditional Pain Point:** Traditional optical logic defines "0 = no light (dark state)" and "1 = light (bright state)". Dark states are physically fragile and 极易被 (easily submerged by) environmental thermal and shot noise, resulting in extremely poor Boolean fidelity.
- **The FIC Definition:** Logic 0 and Logic 1 are both **energy-rich orthogonal bright states**.
 - S_0 (Standard State 0): e.g., pure TE_0 fundamental mode.
 - S_1 (Standard State 1): e.g., pure TE_1 first-order mode, or orthogonal TM_0 polarization state.
- **Physical Significance:** 0 and 1 are completely orthogonal in physical space (mode/polarization). Boolean operations are no longer fragile "destructive interference", but rather **orthogonal projections in phase space**. This endows FIC with an exceptionally high Noise Margin, 彻底消灭 (completely eradicating) the chronic disease of "dark-state noise".

Axiom II: The Axiom of Physical Mapping Configuration (M_C) & Attractor

- Definition:** The fundamental physical unit is defined by the **Physical Mapping Configuration** M_C:
M_C = (G, Mat, B)
where G is the Geometric constraint, Mat is the Material response, and B represents the Boundary conditions.
- Operator Equation:** M_C is not a mere physical entity, but an evolutionary rule in state space. Based on M_C, we define the **Photonic State Constructor operator** PSC_k:
Ψ_{out} = PSC_k[M_C](Ψ_{in})
- Physical Significance:** M_C constructs an "attractor potential well" in phase space. Regardless of how much noise and distortion the input optical field Ψ_{in} carries, once it enters M_C, its boundary conditions B act like a sieve, stripping away spurious modes and forcing the optical field to converge to a pure standard state S_k over a propagation distance L:
lim_{z→L} PSC_k[M_C](Ψ_{in}) = S_k

Axiom III: The Axiom of Self-Healing & Infinite Cascadability

- Mathematical Essence:** Since PSC_k is an operator projecting onto the attractor S_k, it possesses **Idempotence** (PSC(S_k) = S_k) and **Contraction Mapping** properties (|| PSC(Ψ₁) – PSC(Ψ₂) || < || Ψ₁ – Ψ₂ ||).
- Physical Significance:** Minor distortions from the previous stage are forcibly "cleansed" and "reshaped" upon entering the subsequent PSC. Combined with background pump gain (compensating for losses in Mat or B), the FIC system possesses a **Self-Healing** capability akin to a Schmitt trigger in electronics, mathematically and physically guaranteeing **Infinite Cascadability**.

Part II: The Periodic Table of Photonic Elements

Based on M_C and PSC, the output of all elements is standardized to S₀/S₁, achieving standardized interfaces. We abandon traditional "waveguide/coupler" nomenclature and establish a brand-new periodic table based on the physical functions within the FIC architecture.

Group 0: State Reference Elements — The Foundation

Symbol	FIC Name	Physical Mapping Configuration M _C = (G, Mat, B)	Core Function
PSC ₀	Zero-State Constructor	Strip waveguide + Total Internal Reflection (symmetric boundary)	Forcibly converges any input light to Standard State S ₀ (e.g., TE ₀).

Symbol	FIC Name	Physical Mapping Configuration $M_C = (G, Mat, B)$	Core Function
PSC_1	One-State Constructor	Photonic crystal/grating waveguide + Bandgap/Asymmetric boundary	Forcibly converges any input light to Standard State S_1 (e.g., TE_1).

Group 1: Active Control Elements — The Engine of Logic

Symbol	FIC Name	Physical Mechanism (Dynamic M_C)	Core Function (Boolean Embodiment)
T_{opt}	Photonic Transistor (PSC-Gate)	Potential Well Switch. Introduces a control terminal (electrical/thermal/bias light) to dynamically alter Mat or B , instantly switching the attractor well between S_0 and S_1 .	Core switching & amplification. Output S_0 when control is 0, S_1 when 1. Features inherent signal regeneration and gain solving cascading loss.
D_{opt}	Photonic Diode (PSC-Isolator)	Asymmetric Topological Boundary. Utilizes topological photonics to construct a unidirectional B .	Unidirectional routing. Blocks backscattering, protecting upstream logic and ensuring unidirectional signal flow.

Group 2: Passive Tuning Elements — The Voltage Regulators

Symbol	FIC Name	Physical Mechanism (Dissipative/Delay M_C)	Core Function (Impedance & Noise Management)
R_{opt}	Photonic Resistor (PSC-Damper)	Mode Damper. Does not attenuate standard	Smart impedance matching. Filters out-

Symbol	FIC Name	Physical Mechanism (Dissipative/Delay M_C)	Core Function (Impedance & Noise Management)
		states; instead, via controlled sidewall roughness/absorbing materials, it provides immense radiative loss specifically for "non-standard spurious modes".	of-band noise, adjusts optical power levels, and prevents saturation of downstream T_{opt} .
C_{opt}	Photonic Capacitor (PSC-Buffer)	Temporal Potential Well. Utilizes high-Q microring or slow-light photonic crystal structures to increase photon dwell time (group velocity $v_g \ll c$).	Phase buffering & energy storage. Smooths optical pulse jitter, providing precise time delays for all-optical clock synchronization.

Group 3: Logic & Routing Elements — The Building Blocks

Symbol	FIC Name	Physical Mechanism (Composite M_C)	Core Function
L_{opt}	Photonic Logic Gate (PSC-Logic)	Composite Potential Well. Two standard state inputs undergo nonlinear/spatial interference via specific (G, B) , with the result falling into the S_0 or S_1 well.	Boolean operations. Realizes AND/OR/NOT/XOR. Outputs are automatically reshaped into standard states.
Y_{opt}	Photonic Splitter (PSC-Splitter)	Adiabatic Mode Evolution. Maintains the S_k state unchanged while distributing or combining optical	Fan-out. Perfectly copies one S_k into N paths to drive multiple downstream logic gates.

Symbol	FIC Name	Physical Mechanism (Composite M_C)	Core Function
		energy by physical ratios.	

Part III: System Assembly & Infinite Cascading

The core of the FIC architecture is " **M_C Standardization + PSC Pipeline Reshaping**".

1. Paradigm Shift in Logic Gates (Example: FIC-NOT Gate)

- **Traditional Optical NOT Gate:** Relies on destructive interference of two beams. Highly susceptible to temperature fluctuations; fails after 2 stages of cascading.
- **FIC-NOT Gate:**
 - **Input:** Standard State S_0 .
 - **Structure:** A T_{opt} with a control terminal (Dynamic M_C switching).
 - **Operation:** Applying power to the control terminal (Logic 1) alters the local Mat , flipping the potential well of the unit from S_0 to S_1 .
 - **Output:** Pure Standard State S_1 (Logic 0).
 - **Advantage:** The output remains an energy-rich S_1 , capable of directly Fan-out driving 10 downstream FIC-NOT gates.

2. The All-Optical Digital Pipeline

In the FIC substrate, signal flow is no longer an "attenuating wave", but a "continuously reshaped state":

Dirty Input $\rightarrow [PSC_{in}] \rightarrow S_k \rightarrow [L_{opt}] \rightarrow \text{Micro-distortion} \rightarrow [PSC_{reg}] \rightarrow \text{Pure } S_k \rightarrow [T_{opt}] \rightarrow \dots$

Every node is a 2R/3R (Re-shaping + Re-amplifying) regeneration. This is the ultimate solution for FIC to completely shatter the "Optical Wall".

Part IV: Figures of Merit & Physical Anchors

To prevent the theory from floating in pure mathematics, FIC must define measurable physical metrics and provide physical anchors ready for immediate tape-out.

1. Core Figures of Merit (FoM)

- **Convergence Length (L_C):** The physical distance required for an input optical field to evolve from an initial distorted state to a standard state with >99% mode purity (determines device compactness).
- **Mode Extinction Ratio (MER):** The ratio (in dB) of the energy of the standard state S_k to the strongest spurious mode, measuring "Boolean purity".

- **State Tolerance ($\Delta\Psi$):** The maximum allowable distortion range of the input optical field (phase/mode noise margin).

2. First-Generation Physical Anchors (Based on 220nm SOI Platform)

- **M_{C0} Anchor (PSC-0):**
 - *G*: Strip waveguide, width 450nm, height 220nm, length $10\mu m$.
 - *Mat*: Si core ($n \approx 3.48$), SiO₂ cladding.
 - *B*: Symmetric Total Internal Reflection boundary.
 - **Physical Mechanism**: Radiative loss of higher-order modes. Target Standard State: S_0 (TE₀).
- **M_{C1} Anchor (PSC-1):**
 - *G*: Ridge waveguide (width 1200nm) + periodic subwavelength grating (period 300nm), length $20\mu m$.
 - *Mat*: Same as above.
 - *B*: Asymmetric slot coupling boundary + Bragg scattering condition.
 - **Physical Mechanism**: The grating couples and radiates away TE₀; only TE₁ satisfies phase-matching for lossless transmission. Target Standard State: S_1 (TE₁).

Conclusion: The Dawn of Digital Optics

This document, *The FIC Paradigm of Digital Optical All-Optical Computing*, marks a great awakening in photonics.

For the past thirty years, global scientists have been searching for the "perfect nonlinear material" to build optical transistors, a path that has led to a dead end. The FIC architecture points out: **We do not need to wait for God to grant us perfect nonlinear materials. We only need to exquisitely design the Geometry (*G*), Material (*Mat*), and Boundary (*B*) to construct the Physical Mapping Configuration M_C , allowing light to choose which digital state to collapse into on its own.**

First, define the absolute stability of physical digital states ($M_C \rightarrow PSC \rightarrow S_k$); then, assemble logic through the device family.

This is not just the methodology of FIC; it is the physical constitution for the next generation of all-optical digital computing.

FIC begins with the Physical Mapping Configuration.